

ANKIT KUMAR

Department of Management Sciences (DoMS), Indian Institute of Technology Kanpur, UP – 208016, India
ankitk25@iitk.ac.in • [LinkedIn: in/ankit-kumar](#) • [GitHub: Ankitiitk369](#) • [Portfolio: ankitiitk369.github.io](#)

EDUCATION

Indian Institute of Technology Kanpur

Master of Technology (M.Tech), Department of Management Sciences (DoMS)

Cumulative Performance Index (CPI): **7.18 / 10.0**

Kanpur, India

July 2025 – Present

Expected Graduation: June 2027

C.S.A. University of Agriculture and Technology

Bachelor of Technology (B.Tech) in Agricultural Engineering

Cumulative Performance Index (CPI): **8.25 / 10.0**

Kanpur, India

Aug 2021 – June 2025

Awarded **Chancellor's Silver Medal (Rank 1)**

Government Inter College

UP State Board: Class XII – **77.20%** (2021) • Class X – **80.83%** (2019)

Jalaun, UP, India

2018 – 2021

TECHNICAL SKILLS & COMPETENCIES

- **Languages & Core:** Python (Advanced), SQL (PostgreSQL, MySQL), C/C++, HTML5, CSS3, JavaScript, LaTeX
- **Machine Learning & Statistics:** Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Feature Engineering, SMOTE, Hypothesis Testing (A/B Testing, Two-Way ANOVA, Chi-Square), Statsmodels, Time Series (ARIMA, SARIMAX, Prophet)
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, Keras, OpenCV, YOLOv8, EasyOCR, Tesseract, CNN, RNN, Stacked LSTM, Transfer Learning (MobileNetV2, EfficientNetB0)
- **Generative AI & Agentic Systems:** LangChain, LangGraph, LlamaIndex, CrewAI, Groq LLaMA-3.3, Google Gemini 2.5, Pinecone Serverless, FAISS, Qdrant, RAGAS Benchmarks, ReAct Agents, Prompt Engineering, LLMops
- **Analytics, BI & Cloud:** Power BI (Advanced DAX, Power Query), Tableau (LOD, Executive Dashboards), Advanced Excel, FastAPI, Streamlit, Docker, Git, GitHub, Linux, Pytest CI
- **Relevant Coursework (IIT Kanpur):** Applied Machine Learning (AML), Statistical Modelling for Business Analytics (SMBA), Data Mining & Knowledge Discovery, Operations Research for Management, Computing, Supply Chain Management

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Data Science & AI Intern / Computer Vision, OCR & Predictive Modeling

Remote / India

May 2026 – July 2026

- Engineered an automated end-to-end vehicle CO₂ emission tracking pipeline from traffic CCTV video feeds to monitor urban carbon footprints and support enterprise ESG compliance reporting.
- Implemented **YOLOv8** object localization paired with an **OpenCV** preprocessing engine (Bilateral Filtering, CLAHE, Otsu Binarization) and **EasyOCR**, writing a custom regex corrector resolving character optical confusions (O vs 0, I vs 1, B vs 8).
- Integrated Vahan / RTO registry APIs across 36 Indian States/UTs to dynamically extract vehicle metadata (Fuel Type, Engine CC, Vehicle Class) and compute dynamic Vehicle Age from Manufacturing Year.
- Modeled multi-vehicle trip trajectories using Multi-Origin Convergence Routing with Haversine & OSRM matrices for fleet-level carbon density.
- Trained dual-stage ensemble models: **Gradient Boosting ($R^2 = 0.996$, RMSE = 3.12 g/km)** and **Random Forest ($R^2 = 0.989$)** delivered interactive **Streamlit** dashboard with GIS maps and **17-test Pytest** suite (100% pass rate).

Global Procurement Web Application System

Frontend & Asset Automation Contributor

Enterprise IT Project

Feb 2026 – Apr 2026

- Contributed to the Asset Receiving Automation module for an enterprise Global Procurement System, digitizing manual tracking across 100+ categories.
- Engineered responsive **Angular 16** user interfaces for capturing asset metadata, PO numbers, quantities, and physical warehouse storage locations.
- Integrated dynamic form validation schemas, authentication workflows, and **Role-Based Access Control (RBAC)**, cutting asset ingestion latency by **40%**.

KEY PROJECTS: GENERATIVE AI & AUTONOMOUS SYSTEMS

MediMind AI: Clinical Decision Support Platform

Technologies: LangChain, Groq API (LLaMA-3.3-70B), Pinecone Serverless, MiniLM-L6-v2, Streamlit

Nov 2025 – Dec 2025

[\[GitHub Code\]](#)

- Architected an end-to-end RAG clinical diagnostic platform using **LangChain**, **Groq LLaMA-3.3**, and **Pinecone**, cutting inference latency by **85%**.
- Integrated **MiniLM-L6-v2** dense vector embeddings, achieving **96.8% RAG faithfulness** and zero clinical hallucinations with verified citations.
- Automated lab report ingestion via **PyPDF**, embedding 4 validated clinical risk calculators with sub-10ms emergency triage safety guardrails.

KEY PROJECTS: AGENTIC SYSTEMS, MACHINE LEARNING & ANALYTICS

Text-to-SQL & Agentic SQL Analytics System

Jan 2026 – Feb 2026

Technologies: LangGraph, Google Gemini 2.5 Flash, ReAct Agents, MySQL, RAGAS, Plotly

[\[GitHub Code\]](#)

- Developed an autonomous Text-to-SQL system utilizing **Google Gemini 2.5 Flash** and **MySQL** across a 6-table dataset (**64,000+ records**).
- Architected an agentic **LangGraph ReAct** loop using **SQLDatabaseToolkit** with DDL/DML guardrails, attaining **1.00 Context Precision** on **RAGAS**.

Petrol Price Forecasting & Analytics System

Aug 2025 – Nov 2025

Academic Project (Guide: Prof. Faiz Hamid, IIT Kanpur) | PyTorch Stacked LSTM, XGBoost, ARIMA

[\[GitHub Code\]](#)

- Formulated a comparative forecasting pipeline using **ARIMA(5,1,0)**, **XGBoost**, and **PyTorch Stacked LSTM** for weekly pricing.
- Engineered **15+ temporal features** (1–12 lags, SMA/EMA, rolling volatility); tuned XGBoost delivered **1.12% MAPE** and **67.99% directional accuracy** on 35% unseen test data; deployed REST API via **FastAPI**.

Credit Risk Modelling: Loan Eligibility Prediction

Aug 2025 – Nov 2025

Academic Project (Guide: Prof. Faiz Hamid, IIT Kanpur) | Scikit-learn, XGBoost, SMOTE, GridSearchCV

[\[GitHub Code\]](#)

- Constructed an enterprise credit scoring pipeline analyzing **51,336 customer records** with 79 attributes to predict default risk.
- Feature selection via **Chi-Square, ANOVA, and VIF**; handled severe class imbalance via **SMOTE**; achieved **78% test accuracy** via **XGBoost**.

Solar Panel Defect Classification

Jan 2026 – May 2026

Academic Project (Guide: Prof. Veena Bansal, IIT Kanpur) | TensorFlow, MobileNetV2, EfficientNetB0

[\[GitHub Code\]](#)

- Trained a 6-class deep learning defect detection pipeline on **885 electroluminescence images** of photovoltaic cells using **TensorFlow**.
- Applied Batch Normalization, Dropout (0.3), and L2 decay; fine-tuned **MobileNetV2**, attaining **82.5% validation accuracy**; deployed via Streamlit.

Customer Shopping Behavior Analysis & BI Platform

Sep 2025 – Oct 2025

Technologies: PostgreSQL, Power BI, Advanced DAX, Pandas, Relational Modeling

[\[GitHub Code\]](#)

- Analyzed **3,900 consumer purchase records** across 18 demographic/transactional dimensions using Python and relational modeling.
- Formulated complex **PostgreSQL** queries using **CTEs**, window ranking functions, and an executive **Power BI** dashboard with dynamic **DAX measures** and RFM cohorts.

ADDITIONAL SELECTED PRODUCTION & ACADEMIC SYSTEMS

- **Generative AI & Agents:** Multi-Agent Resume Matcher (CrewAI), Multilingual Blog RAG (LangGraph), Dental Clinic State Machine (LangGraph, MySQL), Air India Regulatory Assistant (FAISS, Streamlit), Multi-PDF RAG System, ESG Sustainability Compliance RAG.
- **ML, Vision & Forecasting:** Cancer Risk 3-Class Stratification (>98% acc), Breast Cancer Biopsy FNA (PCA, ROC-AUC), AgriTech Crop Yield Pipeline, Invoice IDP OCR (Tesseract, OpenCV), NVIDIA Stock Forecasting (LSTM, SARIMA), Electricity Demand SARIMAX.
- **Analytics & Business Intelligence:** Loan Portfolio Risk (3 Tableau Dashboards), Zepto Quick-Commerce SQL Analytics, BigMart Sales Multivariate Regression, Multichannel Marketing A/B Testing & Two-Way ANOVA, McDonald's Customer Segmentation (K-Means, PCA).

POSITIONS OF RESPONSIBILITY & LEADERSHIP

- **Alumni and Corporate Relations Coordinator, IIT Kanpur** (July 2025 – Present): Coordinate with corporate leadership and alumni network to drive campus engagement; organized 10+ departmental technical webinars and industrial networking sessions.
- **Campus Coordinator, Mimamsa 2025** (IISER Pune): Led institutional outreach across 15+ universities in northern India; supervised regional round execution.
- **NSS Student Volunteer (2 Years):** Spearheaded grassroots advocacy campaigns for girls' education and digital empowerment under 'Ek Bharat Shreshtha Bharat'.

SCHOLASTIC ACHIEVEMENTS & PROFESSIONAL HONORS

- **GATE 2025:** Secured **98.48 percentile** in Agricultural Engineering, conducted by **Indian Institute of Technology, Roorkee**.
- **Chancellor's Silver Medal Award:** Awarded the prestigious **Silver Medal** for rank 1st academic performance in B.Tech by the University Chancellor.
- **GCI World April 2026:** Global Data Science & AI Program by **Matsuo-Iwasawa Laboratory, The University of Tokyo** (cleared comprehensive evaluation).
- **Professional Certifications:** **Oracle Certified Data Science Professional** • **Oracle Certified AI Foundations** • **SEBI-Investor Certification (NISM)** • **Deloitte Australia Data Analytics** • **British Airways Data Science**.